Tutorial No : 7

Tutorial Title: Evaluation of postfix expression.

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Program:

```
#include <stdio.h>
#include <stdlib.h>
#include <ctype.h>
#include <math.h>
#define MAX 100
double stack[MAX];
int top = -1;
void push(double value)
{   if (top == MAX - 1)
    {      printf("Stack Overflow!\n");
        exit(1);    }
  stack[++top] = value;
} double pop()
{   if (top == -1)
    {     printf("Invalid postfix expression!\n");
        exit(1);   }
   return stack[top--]; }
int main()
{    char postfix[MAX];
    char *p;
    double op1, op2, result;
    printf("Enter postfix expression (single-digit operands): ");
    if (fgets(postfix, sizeof(postfix), stdin) == NULL)
        return 1;
    p = postfix;
   while (*p != '\0' && *p != '\n')
    {      if (isspace((unsigned char)*p))
        {        p++;
          continue;     }
     if (isdigit((unsigned char)*p))
        {        push(*p - '0');
        }        else
```

```
    {       if (*p != '+' && *p != '-' && *p != '*' &&
            *p != '/' && *p != '^')
          {       printf("Invalid character: %c\n", *p);
           return 1;    }
           if (top < 1)
          {          printf("Invalid postfix expression!\n");
             return 1;   }
          op2 = pop();
          op1 = pop();
          switch (*p)
          {       case '+':
                  result = op1 + op2;
                  break;
                   case '-':
                  result = op1 - op2;
                  break;
              case '*':
                  result = op1 * op2;
                  break;
            case '/':
                  if (op2 == 0)
                  {                printf("Division by zero!\n");
                     return 1;          }
               result = op1 / op2;
                  break;
           case '^':
                  result = pow(op1, op2);
                  break;
               default:
                  return 1;          }
           push(result);       }
            p++;   }
 if (top != 0)
  {     printf("Invalid postfix expression!\n");
     return 1;    }
printf("Result = %.2f\n", pop());
 return 0; }
```

Output:

main.c

Run ▶

```c
#include <stdio.h>
#include <stdlib.h>
#include <ctype.h>
#include <math.h>
#define MAX 100
double stack[MAX];
int top = -1;
void push(double value)
{   if (top == MAX - 1)
    {      printf("Stack Overflow!\n");
        exit(1);    }
   stack[++top] = value;
} double pop()
{   if (top == -1)
    {     printf("Invalid postfix expression!\n");
        exit(1);   }
    return stack[top--]; }
int main()
{    char postfix[MAX];
    char *p;
    double op1, op2, result;
    printf("Enter postfix expression (single-digit operands): ");
    if (fgets(postfix, sizeof(postfix), stdin) == NULL)
        return 1;
    p = postfix;
   while (*p != '\0' && *p != '\n')
    {      if (isspace((unsigned char)*p))
        {         p++;
            continue;     }
     if (isdigit((unsigned char)*p))
        {        push(*p - '0');
        }        else
        {          if (*p != '+' && *p != '-' && *p != '*' &&
              *p != '/' && *p != '^')
            {           printf("Invalid character: %c\n", *p);
              return 1;    }
              if (top < 1)
            {             printf("Invalid postfix expression!\n");
                return 1;   }
            op2 = pop();
            op1 = pop();
            switch (*p)
            {         case '+':
                    result = op1 + op2;
                    break;
                     case '-':
                    result = op1 - op2;
```

Output

```
Enter postfix expression (single-digit operands): 23*54*+
Result = 26.00
```